1. Derivatives of Activation Functions

① ReLU

$$f(x) = \begin{cases} x & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$f'(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

* the derivative of ReLU at $x=0$ is supposed to be undefined but I will return 0

② Leaky ReLU

$$f(x) = \begin{cases} x & x \geq 0 \\ ax & x < 0 \end{cases}$$

$$f'(x) = \begin{cases} 1 & x \geq 0 \\ a & x < 0 \end{cases}$$

③ sigmoid function

$$f(x) = \frac{1}{1+e^{-x}}$$

$1+e^{-x} \Rightarrow a$
$e^{-x} \Rightarrow b$
$-x \Rightarrow c$

$$f'(x) = \frac{dy}{da} \times \frac{da}{db} \times \frac{db}{dc} \times \frac{dc}{dx}$$

$$= \frac{1}{(1+e^{-x})^2} \times 1 \times e^{-x} \times -1$$

$$= \frac{e^{-x}}{(1+e^{-x})^2}$$

$$= \frac{1+e^{-x}-1}{(1+e^{-x})^2}$$

$$= \frac{1+e^{-x}}{(1+e^{-x})^2} - \frac{1}{(1+e^{-x})^2}$$

$$= \frac{1}{1+e^{-x}} - \frac{1}{(1+e^{-x})^2}$$

$$= \frac{1}{1+e^{-x}}\left(1 - \frac{1}{1+e^{-x}}\right)$$

$$= f(x)(1-f(x))$$

④ tanh

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{\sinh(x)}{\cosh(x)}$$

$$f'(x) = \frac{\cosh(x) \times \cosh(x) - \sinh(x) \times \sinh(x)}{\cosh^2(x)}$$

$$= \boxed{1 - \tanh^2(x)}$$

* softmax is different from the 4 other activation functions. It will be derived together with CCE